

Abir Ahmed

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Research Interests

Generative modeling for **speech and vision**, focusing on **zero-shot voice cloning**, **speech synthesis**, **diffusion models**, **Flow Matching**, **controllable image generation**, and **multimodal learning**.

Education

BSc. in Software Engineering

Shahjalal University of Science and Technology
2019 – 2024

CGPA: 3.58 / 4.00

Research Experience

Research Assistant – Prof. Mohammad Shahidur Rahman

Department of CSE, Shahjalal University of Science and Technology
September 2024 – July 2025

- Conducted research on **Bengali speech synthesis**, **zero-shot voice cloning**, and **speaker adaptation**.
- Contributed to the work that resulted in the **ICCIT 2024** publication on zero-shot voice cloning.

Research Intern – Advanced Machine Intelligence Research Lab (AMIRL)

April 2024 – August 2024

- Investigated **GANs and diffusion models** for generative image synthesis.
- Fine-tuned **Stable Diffusion** for text-to-image generation and domain-specific synthesis.
- Contributed to the paper **"Percept-Diff"** accepted at IEEE 3ICT 2024.

Research Collaboration – Prof. Mohammad Shahidur Rahman

January 2026 – Present

- Developed **BanglaS2S** dataset; currently investigating direct speech-to-speech translation for Bengali.

Teaching Experience

Adjunct Lecturer – Department of Computer Science & Engineering, Sylhet Engineering College

June 2026 – Present

- Teach undergraduate **Data Structures** and **Structured Programming Language**, including theory and laboratory courses.
- Develop programming exercises, laboratory activities, and assessments.

Publications

Diff-AS: Diffusion-Augmented Strategy for Chrysanthemum Leaf Classification [🔗](#)

Accepted, IEEE QPAIN 2026

- Applied **DreamBooth fine-tuning of Stable Diffusion** using 400 images/class to address data imbalance.
- Combined synthetic and real images and applied transfer learning for classification.

Efficient Zero-Shot Voice Cloning for Bengali Speech Synthesis [🔗](#)

Accepted, ICCIT 2024

- Investigated two approaches for **Bengali zero-shot voice cloning**.
- Proposed a **data-efficient** ensemble approach.
- Achieved MOS scores of **3.8 ± 0.11** and **4.0 ± 0.11** .

Percept-Diff: Innovations in Stable Diffusion for High-Fidelity IHC Image Generation in HER2 Breast Cancer Incorporating Perceptual Loss [🔗](#)

Accepted, 3ICT 2024

- Developed class-conditioned medical image generation using **Stable Diffusion**, **DreamBooth**, and **perceptual loss**.
- Evaluated generation using **CLIP**, **FID**, and **KID**.

Research in Progress.

- Exploring the Barriers Preventing Women in Bangladesh from Pursuing Political Leadership Roles: Dataset Creation and Machine Learning Analysis (Submitted **ACL Workshop 2027**)
- DCT-GAN**: A Diversity-Controlled Training Strategy for High-Diversity Brain Tumor MRI Synthesis – (**First Author**)
- BanglaS2S**: A Large-Scale and Quality-Controlled Bangla–English Speech-to-Speech Parallel Corpus for Speech-to-Speech Translation – (**Second Author**)

Industry Experience

Software Intern – LogiQbits

February 2023 – August 2023

- Developed **LSTM**, **GRU**, and **Random Forest** models for time-series forecasting.
- Retrieved, analyzed, and visualized sales data from large databases.

Technical Skills

Programming

Python, PyTorch, C, C++, Java, Javascript, TypeScript

Speech & Audio:

Speech Synthesis, Speech Translation, Speaker Embeddings, TTS

Generative & Multimodal AI

Diffusion Models, GANs, Flow Matching, LLMs, VLMs

Languages

Bangla (Native)

English (IELTS 6.5 Overall)